# Lab 1 – Smart City

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# 1 Introduction

Cities are getting bigger and more crowded every year. The United Nations projects that by 2025, approximately 6.5 billion people will be living in urban areas (Luoma, 2016). This represents nearly 68% of the world's population concentrated in cities. Because of this rapid urbanization, people deal with longer commutes, increased traffic congestion, higher pollution levels and reduced quality of life. The way most cities are currently designed cannot handle this kind of growth, so problems like traffic jams, overcrowding at popular destinations, and poor resource management continue to worsen (Joyce, 2025).

The traditional approach to urban planning has relied on building more roads, expanding public transit systems, and constructing new infrastructure to accommodate population growth. However, this methodology can no longer keep pace with this rate at which cities are expanding (Weinburg, 2016). Construction projects take years to complete and require substantial public funding that many cities struggle to secure. Current city management systems operate largely in isolation, with minimal real-time communication between residents, transportation networks, and municipal planning departments.

The problems urban residents face are considerable. Long commute times, lack of real-time information about local conditions, limited communication channels for reporting issues, and insufficient tools for discovering alternative destinations are significant issues that reduce quality of life (Lee, 2025). Because of these issues, effective information sharing and good communication between city residents and municipal planners are essential. Currently, there is a

lack of tools available that support both effective and efficient communication and the exchange of real-time urban information among all stakeholders.

Existing navigation applications, phone calls to city services, and social media posts are the primary ways that residents communicate about local conditions and receive information about their communities. These communication methods can be inefficient and may have long turnaround times. When residents attempt to report issues like traffic accidents, road closures, or overcrowding at popular destinations, this process typically requires calling municipal hotlines or posting on various disconnected social media platforms. This is usually more time consuming than a centralized reporting system and is ill-suited for tasks like providing location-specific updates or receiving real-time notifications about conditions along planned routes. These issues, collectively, describe a need for a centralized platform that allows residents to share information, navigate efficiently, and communicate about local conditions in real time.

Smart City is an application designed to address urban overcrowding by connecting residents to real-time community information and intelligent navigation tools. This application provides users and city officials with tools that improve efficiency and accessibility to local information, thereby allowing both residents and planners to make better use of available data. The platform offers connectivity, as Smart City creates a network where residents can share and access hyperlocal updates. It provides intelligence, because the system uses real-time data to generate smart routing suggestions. The application ensures community engagement, so residents can actively contribute to improving their urban environment through shared observations. Finally, it delivers clarity, because both residents and city officials will have access to timely, accurate information about local conditions and infrastructure needs.

# 2 Product Description

Smart City is a cross-platform mobile application designed to provide citizens with real-time urban information while enabling city planners and officials to collect valuable data for informed decision-making. The application addresses the growing challenges of urban congestion and overcrowding by combining navigation assistance, crowd monitoring, and intelligent route optimization with a location-based community forum into a unified platform.

## 2.1 Key Product Features and Capabilities

The key features of Smart City include its real-time traffic and crowd monitoring capabilities that provide users with live updates about conditions in their area; an Alternative Destination system that suggests less crowded venues when popular locations reach capacity; integration with mapping services through the Mapbox API for accurate navigation and routing; and a location-based community forum where residents can share updates, report local issues, and engage in neighborhood discussions. The application analyzes traffic patterns and congestion data, while its data collection and analytics framework enables city officials to identify infrastructure bottlenecks and make evidence-based planning decisions.

User features enable citizens to navigate their communities more efficiently by receiving personalized route suggestions based on real-time conditions. The application's Alternative Destination feature helps users discover alternative venues when their intended destination is crowded, reducing frustration and wait times. Users can access the community forum to share updates about local conditions, report issues, and stay informed about their neighborhood. They receive push notifications about traffic conditions, crowd levels, suggested routes, and forum

activity, all while maintaining privacy through secure, anonymized data collection practices. The forum provides thread categories, search functionality, and moderation tools to ensure meaningful community engagement.

City officials and planners benefit from Smart City's data collection capabilities that aggregate user information to reveal traffic patterns, popular destinations, and infrastructure stress points. The platform collects data on urban flow in real-time, enabling officials to identify areas requiring infrastructure improvements and measure the impact of urban planning decisions. By analyzing collected data, officials can implement targeted changes to roadways, adjust traffic management systems, and optimize public transportation routes to better serve community needs.

## 2.2 Major Components (Hardware/Software)

The user mobile application is compatible with both Android and iOS devices. Users interact with the mobile application to receive navigation guidance, view real-time traffic and crowd data, access the community forum, and post updates about local conditions. The data they generate is collected securely and transmitted to the backend system for storage and processing.

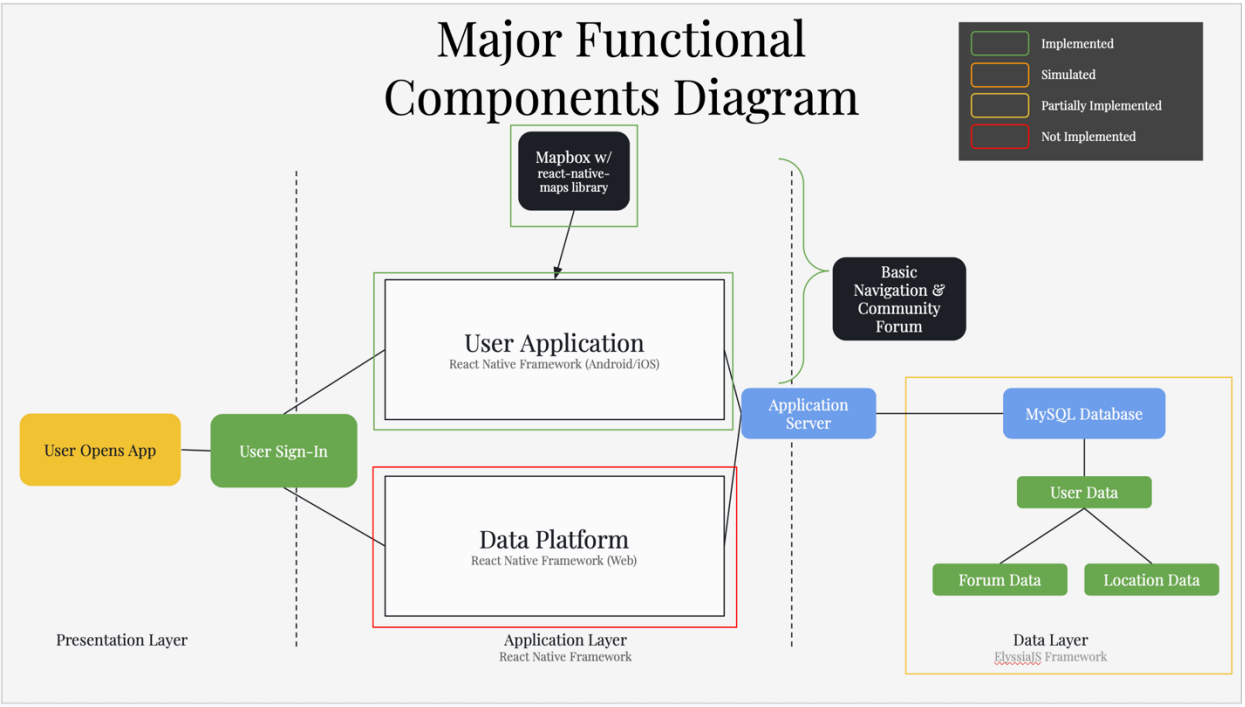


**Figure 1 - Major Function Component Diagram**

The Major Functional Components Diagram is split into three different sections. We first have the presentation layer which allows users and city officials to engage with the application. Then we have the application layer which provides the logical components that make up the application. This includes the core data handling logic. Lastly, we have the data layer which houses the database we will use for data storage.

Smart City incorporates several key software technologies: React Native serves as the framework for the cross-platform mobile application, with Expo providing the development environment and toolchain for building, deploying, and rapidly iterating on the application across iOS and Android devices. Mapbox API integration provides all mapping, navigation, and routing capabilities within the user application, including real-time traffic visualization and turn-by-turn directions. JavaScript is used as the primary programming language for development.

The backend infrastructure utilizes Bun as the JavaScript runtime environment, providing high-performance server-side execution and efficient handling of concurrent user requests. The MySQL database, hosted on Old Dominion University's servers, stores user data, forum posts and comments, traffic information, and analytics results. Firebase Authentication handles user sign-in with support for Google and Apple sign-in options. Push notifications keep users informed of traffic conditions, overcrowding alerts, and forum activity.

## 3 Identification of Case Study

The main users for Smart City are urban residents, daily commuters, and city officials responsible for urban planning. Rapid urbanization and poor planning create overcrowding in metropolitan areas, reducing quality of life through longer commutes and traffic congestion. Smart City addresses the lack of centralized, real-time information by allowing citizens to navigate efficiently while providing city planners with data for informed infrastructure decisions.

Urban residents and commuters face significant challenges navigating congested cities with limited real-time information. Smart City provides personalized route suggestions, alternative destination recommendations when locations are crowded, and a community forum for sharing local updates. These features give users control over their daily commutes and enable informed decisions about when and where to travel.

City officials need reliable data about traffic patterns and infrastructure stress points to make evidence-based planning decisions. Smart City provides aggregated, anonymized data from user activity, revealing insights about urban flow and community needs that would be difficult to obtain through traditional methods.

The purpose of Smart City is to reduce urban congestion, improve navigation efficiency, and provide usable data for urban planning. To test these goals, a case study will be performed using five to ten Old Dominion University students and nearby residents as a beta group. Participants will use Smart City as their primary navigation tool and engage with the community forum to share updates about local conditions. The beta group will provide ongoing feedback on usability, route accuracy, and forum usefulness. An evaluation will be conducted at the end of the study period, and data collected through the analytics system will be reviewed to assess traffic pattern identification, user engagement, and the effectiveness of alternative destination suggestions. The target group represents typical urban users who face daily navigation challenges in congested areas.

As more users and communities adopt Smart City, their feedback will be essential for improvement and expansion. The platform could scale beyond Norfolk to serve the broader Hampton Roads region. Local businesses could share events while gaining insights about community engagement. The application could be customized for different urban areas and expanded to include public transportation integration, real-time parking availability, and enhanced analytics for city planners.

# 4 Glossary

**Internet of Things (IoT):** systems that combine sensing, computing, communication, and actuation to connect physical/cyber objects

**Application Programming Interface (API):** a set of features and rules that lets software interact with other software (a "contract" between programs)

**Crowdsourced Reporting:** gathering information by soliciting contributions from a large group of people, typically via the internet

**Privacy-by-Design ("Privacy-First Design"):** building systems so that, by design and by default, only the personal data necessary for a purpose are processed

**Open Data (Open Data Access):** data anyone can freely use, re-use, and redistribute, with at most attribution/share-alike requirements

**Predictive AI / AI-Powered Forecasting:** using data, statistical algorithms, and ML to estimate the likelihood of future outcomes

**Anomaly Detection:** identifying data patterns that deviate from expected behavior

**Access Control:** controls that enforce authorized access to information and system resources in line with policy

**Citizen Feedback Loop:** a two-way process where institutions act on and report back about citizen input to sustain engagement and trust

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